

# Cooling Cost Curves in Commercial Complexes: U.S. & China Case Study

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LVMH



恒隆地產  
HANG LUNG PROPERTIES

# Introduction & Motivation

Cooling as a global challenge

- Rapid urban commercial growth, especially in emerging markets (e.g., China)
- Half of global population in hot climates + global warming → rising demand.
- Today: 2 billion ACs globally → projected 5.5 billion by 2050 (IEA).
- Cooling already  $\approx 1/6$  of building electricity use, fastest-growing end-use

Refrigerants Matter

- Core working fluid; annual leakage rate 15-25%
- High-GWP, often untracked and unregulated.

Information Gap

- Opaque data on stock, use, and emissions.
- Large real estate firms (e.g., Hang Lung) and tenants (LVMH) lack reporting.
- Mitigation needs vary by climate zone and building type.

# Project Scope & Objectives

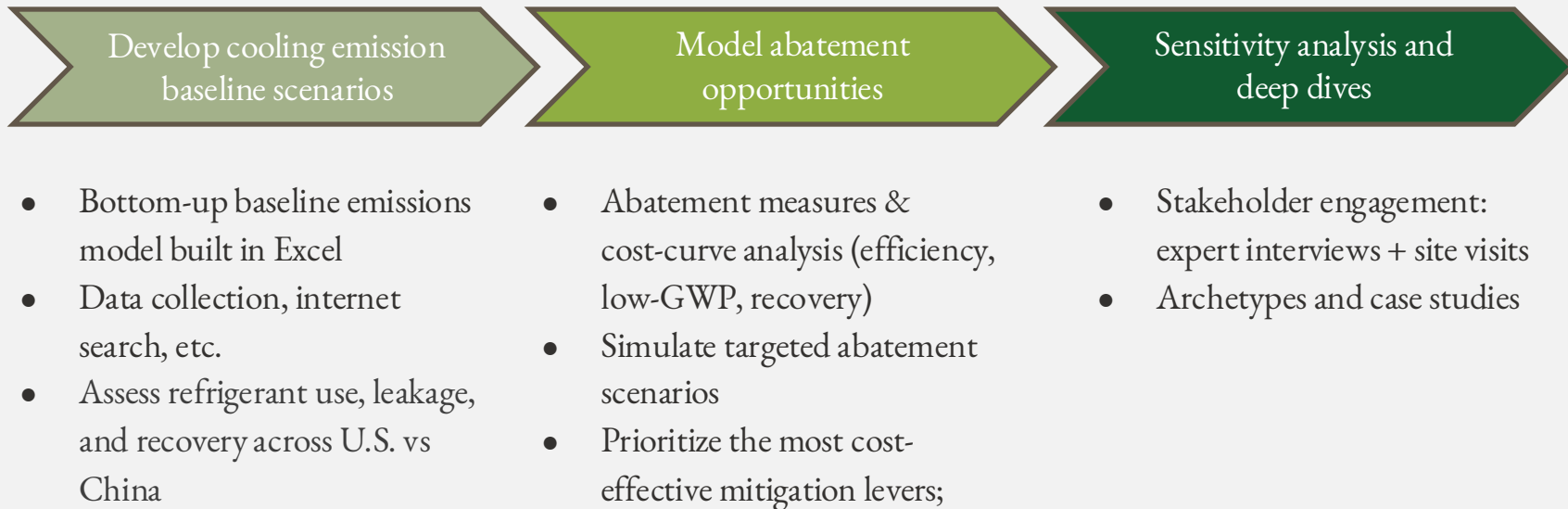


## Focus

- **Sector:** large-scale commercial buildings (e.g., malls, offices, retail)
- **Time scope:** 2024-2060
- **2 markets:** US & China
- **2 categories:** AC + Refrigeration
- **2 emissions:** refrigerant leaks/end of life venting (Scope 1) + cooling energy use (Scope 2)

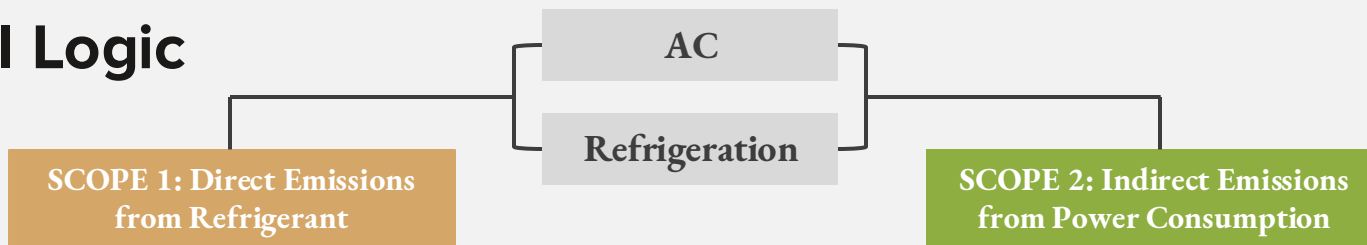
***Objective:** Build business case & cost curve for reducing cooling-related GHG emissions, focusing on refrigerant management.*

# Methodology



More circular than linear!

# Model Logic



Scope 1 =  $\sum \text{by tech \& refrigerant} [\text{Charge} \times (\text{Leak Rate} + \text{EOL Vent Rate}) \times \text{GWP}]$

Scope 2 =  $(\text{Cooling demand} \div \text{Weighted COP}) \times \text{Grid factor}$

**Refrigerant Volume by R-Type =**

Cooling tech/R mix  $\times$  Typical refrigerant charge per tech

**Annual leakage & venting** = R volume  $\times$  (Leak rate + EOL Vent Rate, by tech and R-type)

**Direct emissions =**

Annual leaked/vented R mass  $\times$  GWP

**Cooling Demand =**

$f(\text{floor area} \times \text{cooling energy intensity (kWh/m}^2))$

**Power consumption =**

Cooling demand  $\div$  Weighted SCOP

**Indirect emissions =**

Power consumption  $\times$  Grid GHG intensity

**Total baseline annual cooling emissions**

= (HVAC + Refrigeration)  $\times$  (SCOPE 1 from refrigerant emission + SCOPE 2 from electricity use)

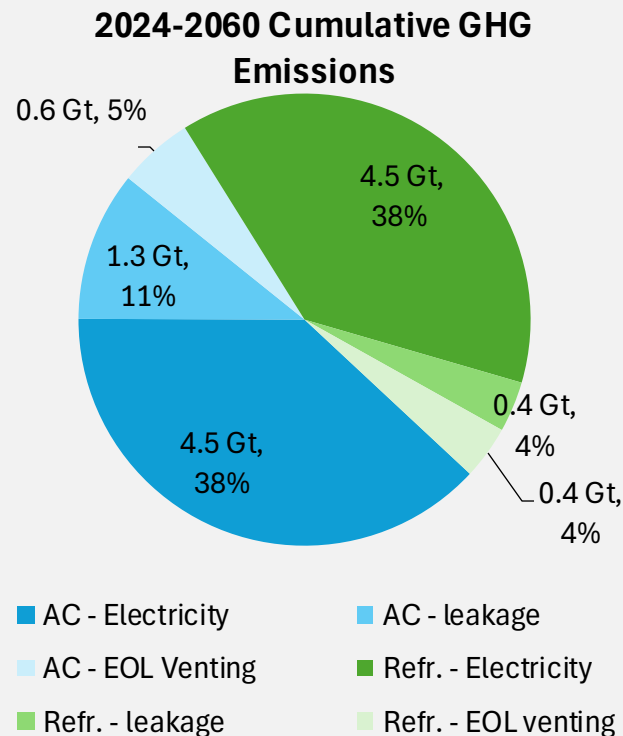
|    | A | B                                       | C       | D     | E     | F     | G     | H     | I     | J     | K     | L     | M     | N     | O     | P     | Q     | R     | S     | T     | U     | V     | W     | X     | Y     | Z     | AA     | AB    | AC    | AD    | AE    | AF    | AG    |     |
|----|---|-----------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|-------|-------|-------|-------|-----|
| 1  |   |                                         |         | TOTAL | 2024  | 2025  | 2026  | 2027  | 2028  | 2029  | 2030  | 2031  | 2032  | 2033  | 2034  | 2035  | 2036  | 2037  | 2038  | 2039  | 2040  | 2041  | 2042  | 2043  | 2044  | 2045  | 2046   | 2047  | 2048  | 2049  | 2050  | 2051  | 2052  |     |
| 2  |   |                                         |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 3  |   | Commercial AC                           |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 4  |   | Electricity                             |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 5  |   | Cooling demand                          | Twh     |       | 1,150 | 1,207 | 1,268 | 1,331 | 1,397 | 1,467 | 1,541 | 1,618 | 1,699 | 1,784 | 1,873 | 1,966 | 2,065 | 2,168 | 2,276 | 2,390 | 2,510 | 2,585 | 2,662 | 2,742 | 2,825 | 2,909 | 2,997  | 3,086 | 3,179 | 3,274 | 3,373 | 3,474 | 3,578 |     |
| 6  |   | Cooling demand                          | CAGR    |       |       | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 3%    | 3%    | 3%    | 3%    | 3%    | 3%     | 3%    | 3%    | 3%    | 3%    | 3%    | 3%    |     |
| 7  |   | Weighted SCOP                           |         |       | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2    | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   | 5.2   |     |
| 8  |   | Power Consumption                       | Twh     |       | 222   | 233   | 245   | 257   | 270   | 283   | 297   | 312   | 328   | 344   | 361   | 379   | 398   | 418   | 439   | 461   | 484   | 499   | 514   | 529   | 545   | 561   | 578    | 596   | 613   | 632   | 651   | 670   | 690   |     |
| 9  |   | Grid GHG intensity (weighted)           | kg/kwh  |       | 0.6   | 0.58  | 0.57  | 0.55  | 0.53  | 0.52  | 0.50  | 0.48  | 0.47  | 0.45  | 0.43  | 0.42  | 0.40  | 0.38  | 0.37  | 0.35  | 0.33  | 0.32  | 0.30  | 0.28  | 0.27  | 0.25  | 0.23   | 0.22  | 0.20  | 0.18  | 0.17  | 0.15  | 0.13  |     |
| 10 |   | Total emissions                         | MT CO2  | 4,468 | 133   | 136   | 139   | 141   | 144   | 146   | 149   | 151   | 153   | 155   | 157   | 158   | 159   | 160   | 161   | 161   | 161   | 158   | 154   | 150   | 145   | 140   | 135    | 129   | 123   | 116   | 108   | 101   | 92    |     |
| 11 |   | Refrigerant leaks                       |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 12 |   | R capacity                              | KW/kg   |       | 2.46  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 13 |   | Weighted utilization                    |         |       | 27%   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 14 |   | Commercial AC volume                    | kTons   |       | 195   | 204   | 215   | 225   | 237   | 248   | 261   | 274   | 288   | 302   | 317   | 333   | 350   | 367   | 385   | 405   | 425   | 438   | 451   | 464   | 478   | 493   | 507    | 523   | 538   | 554   | 571   | 588   | 606   |     |
| 15 |   | Commercial leak rates                   | %       |       | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    | 9%     | 9%    | 9%    | 9%    | 9%    | 9%    | 9%    |     |
| 16 |   | Total annual leak                       | MT CO2e | 1,256 | 41.9  | 43.4  | 44.8  | 46.2  | 47.6  | 48.8  | 49.9  | 50.9  | 51.7  | 52.4  | 52.8  | 53.1  | 53.0  | 52.5  | 51.5  | 51.1  | 50.1  | 47.8  | 45.3  | 42.6  | 39.7  | 36.5  | 33.3   | 30.1  | 26.8  | 23.5  | 20.2  | 16.9  | 13.9  |     |
| 17 |   | Collection rate                         |         |       | 22%   | 23%   | 23%   | 24%   | 24%   | 25%   | 25%   | 26%   | 27%   | 27%   | 28%   | 28%   | 29%   | 29%   | 30%   | 30%   | 31%   | 31%   | 32%   | 32%   | 33%   | 33%   | 34%    | 34%   | 35%   | 35%   | 36%   | 36%   | 36%   |     |
| 18 |   | End of life net emission rate           |         |       | 2.4%  | 2.3%  | 2.3%  | 2.3%  | 2.3%  | 2.3%  | 2.3%  | 2.3%  | 2.2%  | 2.2%  | 2.2%  | 2.2%  | 2.2%  | 2.2%  | 2.2%  | 2.1%  | 2.1%  | 2.1%  | 2.1%  | 2.1%  | 2.1%  | 2.0%  | 2.0%   | 2.0%  | 2.0%  | 2.0%  | 2.0%  | 2.0%  | 1.9%  |     |
| 19 |   | End of life leak                        | MT CO2e | 628   | 10.6  | 11.0  | 11.5  | 12.0  | 12.5  | 13.0  | 13.6  | 14.2  | 14.8  | 15.4  | 16.1  | 16.8  | 17.5  | 18.3  | 19.0  | 19.8  | 20.7  | 20.9  | 21.0  | 21.1  | 21.1  | 21.1  | 21.0   | 20.9  | 20.4  | 20.0  | 19.6  | 19.0  |       |     |
| 20 |   |                                         |         |       | 52.4  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 21 |   | Commercial Refrigeration                |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 22 |   | Electricity                             |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 23 |   | Cooling demand                          | Twh     |       | 223   | 235   | 246   | 259   | 272   | 285   | 299   | 314   | 330   | 347   | 364   | 382   | 401   | 421   | 442   | 464   | 488   | 502   | 517   | 533   | 549   | 565   | 582    | 600   | 618   | 636   | 655   | 675   | 695   |     |
| 24 |   | Refrigeration demand                    | CAGR    |       |       | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 5%    | 3%    | 3%    | 3%    | 3%    | 3%    | 3%     | 3%    | 3%    | 3%    | 3%    | 3%    | 3%    |     |
| 25 |   | Grid GHG intensity (weighted)           | kg/kwh  |       | 0.6   | 0.6   | 0.6   | 0.6   | 0.5   | 0.5   | 0.5   | 0.5   | 0.5   | 0.5   | 0.4   | 0.4   | 0.4   | 0.4   | 0.4   | 0.4   | 0.3   | 0.3   | 0.3   | 0.3   | 0.3   | 0.2   | 0.2    | 0.2   | 0.2   | 0.2   | 0.2   | 0.1   |       |     |
| 26 |   | Total emissions                         | MT CO2  | 4,500 | 134   | 137   | 140   | 142   | 145   | 147   | 150   | 152   | 154   | 156   | 158   | 159   | 160   | 161   | 162   | 163   | 163   | 159   | 155   | 151   | 146   | 141   | 136    | 130   | 124   | 117   | 109   | 101   | 93    |     |
| 27 |   | Refrigerant leaks                       |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 28 |   | R capacity                              | KW/kg   |       | 1.50  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 29 |   | Weighted SCOP                           |         |       | 3.9   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 30 |   | Weighted utilization                    |         |       | 75%   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 31 |   | R volume                                | kTons   |       | 89    | 93    | 98    | 103   | 108   | 113   | 119   | 125   | 131   | 138   | 145   | 152   | 159   | 167   | 176   | 185   | 194   | 200   | 206   | 212   | 218   | 225   | 231    | 238   | 246   | 253   | 261   | 268   | 276   |     |
| 32 |   | Commercial leak rates                   | %       |       | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   | 17%    | 17%   | 17%   | 17%   | 17%   | 17%   | 17%   |     |
| 33 |   | Total annual leak                       | MT CO2e | 424   | 22.5  | 22.4  | 22.2  | 22.0  | 21.6  | 21.2  | 21.2  | 21.3  | 21.2  | 21.1  | 21.0  | 20.7  | 19.7  | 18.5  | 17.1  | 15.5  | 13.7  | 12.1  | 10.4  | 8.6   | 6.7   | 4.6   | 4.4    | 4.2   | 4.0   | 3.8   | 3.6   | 3.3   | 3.1   |     |
| 34 |   | Collection rate                         |         |       | 8%    | 9%    | 9%    | 10%   | 10%   | 11%   | 11%   | 12%   | 12%   | 13%   | 13%   | 14%   | 14%   | 15%   | 15%   | 16%   | 16%   | 17%   | 17%   | 18%   | 18%   | 19%   | 19%    | 20%   | 21%   | 21%   | 22%   | 22%   |       |     |
| 35 |   | End of life net emission rate           |         |       | 4.5%  | 4.5%  | 4.4%  | 4.4%  | 4.4%  | 4.4%  | 4.3%  | 4.3%  | 4.3%  | 4.2%  | 4.2%  | 4.2%  | 4.2%  | 4.2%  | 4.1%  | 4.1%  | 4.1%  | 4.1%  | 4.0%  | 4.0%  | 4.0%  | 4.0%  | 3.9%   | 3.9%  | 3.9%  | 3.9%  | 3.9%  | 3.8%  | 3.8%  |     |
| 36 |   | End of life leak                        | MT CO2e | 449   | 10.8  | 11.3  | 11.8  | 12.3  | 12.9  | 13.4  | 14.0  | 14.7  | 15.3  | 16.0  | 16.7  | 17.4  | 18.2  | 18.1  | 18.0  | 17.7  | 17.4  | 16.6  | 15.7  | 14.8  | 13.7  | 12.5  | 11.3   | 10.9  | 10.5  | 10.1  | 9.7   | 9.2   | 8.7   |     |
| 37 |   |                                         |         |       | 33.3  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 38 |   | Drivers                                 |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 39 |   | Annual GWP for newly installed / refill |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | 2,990 | 199.34 |       |       |       |       |       |       |     |
| 40 |   | AC Residential & small commercial       |         |       | 1,897 | 1,513 | 1,473 | 1,434 | 1,394 | 1,355 | 1,315 | 1,280 | 1,246 | 1,211 | 1,176 | 1,141 | 1,107 | 1,037 | 967   | 897   | 827   | 757   | 673   | 590   | 507   | 423   | 340    | 313   | 286   | 259   | 232   | 205   | 185   | 164 |
| 41 |   | AC Large Commercial                     |         |       | 2,320 | 1,925 | 1,846 | 1,767 | 1,688 | 1,609 | 1,530 | 1,460 | 1,391 | 1,321 | 1,252 | 1,182 | 1,113 | 988   | 862   | 737   | 612   | 487   | 397   | 307   | 217   | 126   | 36     | 29    | 23    | 16    | 9     | 3     | 3     | 3   |
| 42 |   | Commercial Refrigeration                |         |       | 2,827 | 1,503 | 1,424 | 1,345 | 1,266 | 1,187 | 1,108 | 1,058 | 1,009 | 959   | 909   | 859   | 809   | 731   | 654   | 576   | 498   | 420   | 360   | 300   | 241   | 181   | 121    | 113   | 105   | 97    | 89    | 82    | 74    | 66  |
| 43 |   | Residual years                          |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 44 |   | AC Residential & small commercial       |         |       | 15    | 14    | 13    | 12    | 11    | 10    | 9     | 8     | 7     | 6     | 5     | 4     | 3     | 2     | 1     | -     | -     | -     | -     | -     | -     | -     | -      | -     | -     | -     | -     | -     | -     | -   |
| 45 |   | AC Large Commercial                     |         |       | 16    | 15    | 14    | 13    | 12    | 11    | 10    | 9     | 8     | 7     | 6     | 5     | 4     | 3     | 2     | 1     | -     | -     | -     | -     | -     | -     | -      | -     | -     | -     | -     | -     | -     | -   |
| 46 |   | Commercial Refrigeration                |         |       | 12    | 11    | 10    | 9     | 8     | 7     | 6     | 5     | 4     | 3     | 2     | 1     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -      | -     | -     | -     | -     | -     | -     | -   |
| 47 |   | Stock GWP                               |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |
| 48 |   | AC Residential & small commercial       |         |       | 1,871 | 1,843 | 1,812 | 1,778 | 1,742 | 1,704 | 1,662 | 1,619 | 1,573 | 1,525 | 1,475 | 1,422 | 1,365 | 1,303 | 1,236 | 1,190 | 1,143 | 1,092 | 1,038 | 982   | 922   | 860   | 798    | 736   | 675   | 614   | 554   | 497   | 444   |     |
| 49 |   |                                         |         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |     |

## Preliminary Results for China

- 2024 Baseline Emissions
  - Power footprint similar for AC & Refr. (133 Mt CO<sub>2</sub>e)
  - AC has higher refrigerant emission footprint (50 Mt CO<sub>2</sub>e) than Refr. (30 Mt CO<sub>2</sub>e)
- 2024-2060 Cumulative Emissions
  - AC: 6.4 Gt CO<sub>2</sub>e
  - Refrigeration: 5.4 Gt CO<sub>2</sub>e

Question:

Why Refr. venting  $\approx$  leaks, but AC venting  $< \frac{1}{2}$  leaking?



# Stakeholder Interviews

China (Willem): first round completed  
U.S.: 4 interviewed, more scheduled.

- Technology suppliers
- Facility managers
- Users / owners
- Research/ data
- Consultants / private sector
- Non-profit / associations...





## U.S. Interview Insight Highlights

### Regulatory Landscape

- Tightened AIM Act rules (15 lb threshold) pull mid-size commercial units into compliance.
- State EE programs (esp. CA) now reward both energy & refrigerant savings.

- Even large institutions (Yale, Tishman Speyer, Hang Lung) struggle to track inventories.
- Rely on ad-hoc spreadsheets or technician notes; Trakref adoption slow.
- Misalignment among building owners, property managers & outsourced maintenance.

### Data Visibility & Gap

### Behavior & Incentives

- Reactive maintenance over proactive refrigerant management
- OEMs slow-walk low-GWP/natural refrigerants to avoid cannibalizing existing sales.

- Commercial refrigeration is ahead (CO<sub>2</sub> stores, propane cases); HVAC lagging.
- ACTs exist but lack OEM scale-up.
- Reclaimed vs. virgin gas: awareness and supply chain logistics are the bottlenecks.

### Technology Readiness

### Trade-offs

- Environmental impact vs day-to-day business operations
- Energy efficiency vs refrigerant management: chillers with pressure monitoring leak far less than distributed units but are less energy efficient.

# Abatement Measures List

## Cooling Generation Technologies

### Alternative and Emerging Cooling

#### Active cooling:

- Elastocaloric
- Magnetocaloric
- Thermoelectric
- Barocaloric
- Thermoacoustic
- Air-based Cooling

#### Hybrid systems

- Dessicant-based
- Phase-change materials
- Ocean, lake, groundwater cooling

### Heat Pump & Water Loop Innovations

- Larger delivery capacity to reduce temperature differentials
- Waterfalls, capillary systems
- Replace VRF with secondary loop or water loop systems

## Policy and Regulatory Mechanisms

### Efficiency Standards and Certification

- LEED / Star labeling systems for cooling efficiency
- Mandatory refrigerant tracking and reporting

### Carbon Market & Incentive Alignment

- Verified carbon standards
- Incentives tied to refrigerant recovery and system upgrades

## Refrigerant Management

### Leak Detection and Prevention

- Preventive maintenance
  - Use of low-permeability hoses and fittings
  - Ensuring proper installation and pipe insulation
  - Regular leak testing (e.g., with nitrogen or electronic)
- Advanced sensing and monitoring
- Leak detection standards

### Refrigerant Lifecycle Management

- Conduct refrigerant inventories
  - Logging refrigerant type and quantity per system
  - Centralized digital tracking system for refrigerant stock

#### Increase recovery of refrigerants from at end-of life

- Require certified technicians to recover refrigerants
- Incentivize returns through deposit-refund systems
- Use recovery equipment during servicing and dismantling
- Reclaim old refrigerants
- Create centralized reclaim and redistribution centers
- Mandate reclaimed refrigerant quotas for service providers

#### Increasing destruction / disposal

- Partner with certified destruction facilities (e.g., plasma)
- Offer incentives or mandates for destruction over venting
- Establish logistics chain for transport of collected refrigerant
- End-of-life collection and recycling (especially large-scale systems)

### Transition to Low-GWP Refrigerants

- Drop-in low-GWP refrigerants
- Faster rollout of new refrigerants
  - Streamline regulatory approvals for A2L/A3 refrigerants
  - Subsidize or incentivize early adopters (e.g., rebates on equipment)
- Natural refrigerants

## Behavior, Incentives, and Operational Practices

## Example

### Operator Practices and Training

- Better training and expertise of operators
- Technician training and workforce development
- Regular inspection, calibration, and maintenance

[EPA Section 608](#)

### Smart Operation and Control

- More intelligent operation / analysis
- Gradual cooling ramp-up / variable set-points
- Higher cooling set-points
- Smart controls and scheduling
- District cooling systems

[Cooling Singapore](#)

### Tenant Engagement and Accountability

- Separate energy charges for tenants
- Incentives/limitations for tenant-installed cooling tech
- Annual performance targets for building managers

### Data and Measurement

- Better measurement and real-time monitoring
- Software-enabled inventory compilation

[Fexa TrakRef, FMHero](#)

## Cooling System Retrofit and Design

### HVAC System Upgrades

- Combine cooling with hot water/heat delivery
- Desiccant at air inlet
- Variable water/air flow and valve controls
- Cleaning of heat exchangers / air filters
- Isolation of system components

replace gas, expand heat pump use  
[Blue Frontier, Mojave HVAC](#)

insulated pipes

### Heat Exchange and Passive Cooling

- In/out air heat exchange

- Air screens at exits/inlets

i.e. air curtain systems or air doors installed at entrances/exits of buildings or cold storage areas.

#### Passive cooling

- Sky radiative cooling
- Two-phase cooling
- Advanced building materials

[SkyCool Systems, ChillSkyn](#)  
[Calyos](#)  
[Kéré Architecture](#)

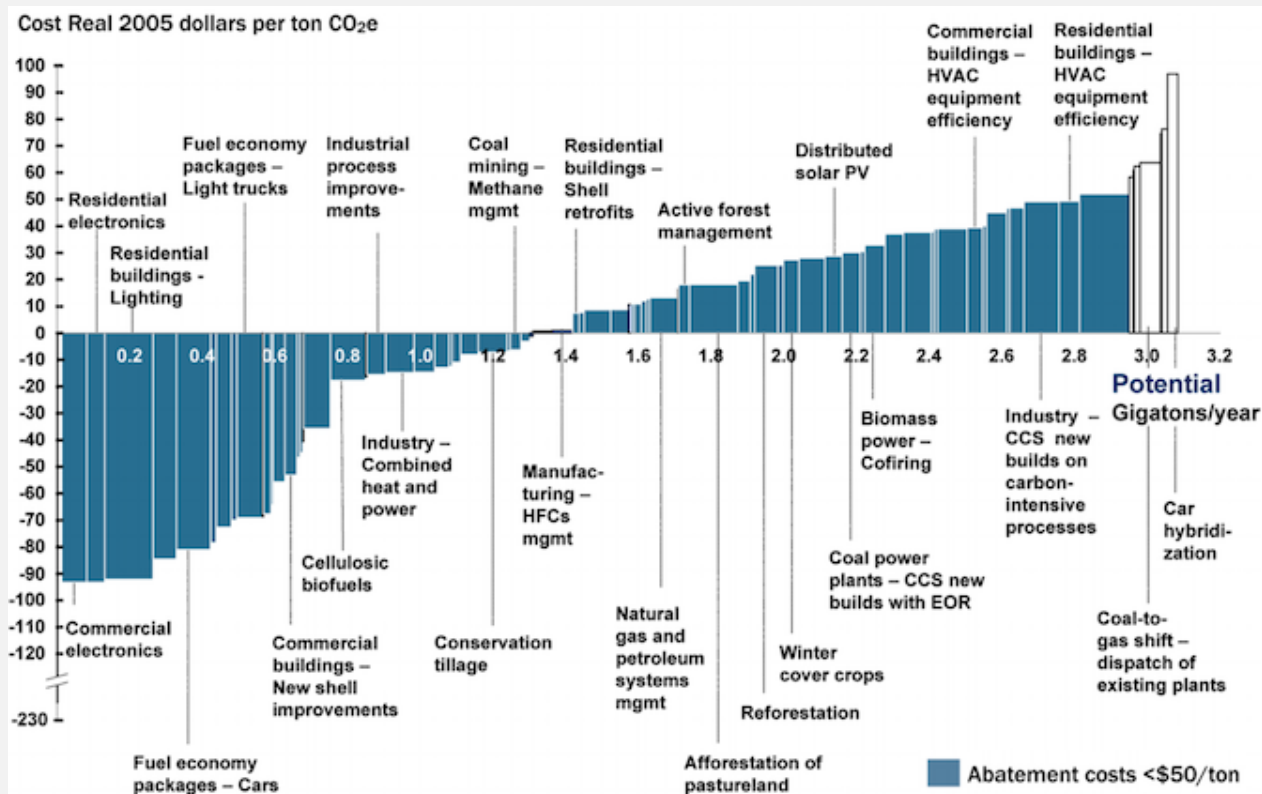
### Building Envelope Improvements

- Shading, trees, foil for sun-facing facades
- Separate hard-to-cool areas

# Final Output: Cost Curve

**Y-axis:** marginal cost of abating 1 t CO<sub>2</sub>e

**X-axis:** cumulative-to-target-year abatement potential—the quantity of CO<sub>2</sub>e you can avoid if the measure is rolled out across the selected stock



## Next Steps

- Wrap up baseline analysis for US model (Aug 11th)
- Integrate abatement measures to the cost curve (late Aug)
- Synthesis findings and draft report (by mid Sep)
- Stay cool 😎

